Lecture 9

$Ax = 0$ for $m > n$ (more unknowns than equations)

then non zero x would be solutions.

⇒ Linear Independence

Vectors $v_1, v_2 \ldots v_n$ are linearly independent if no linear combination except all zeros give 0.

$$\Rightarrow \quad a_1 v_1 + a_2 v_2 \ldots a_n v_n = 0$$

when all $a_i \neq 0$ then vectors are linearly dependent.

If any of these vectors is the zero vector then the vectors are definitely dependent.

eg $V_1 = 0$

$$aV_1 + 0V_2 + 0V_3 \cdots + 0V_n = \underline{0}$$

$$\forall a \in \mathbb{R} \mid \mathbb{C}$$

If there is anything in the nullspace other than the 0 vector, then the vectors are dependent

$$A\underset{\sim}{x} = 0$$

$\hookrightarrow x \neq 0$ then there is a linear combination of $C(A)$ for which it is 0

Rank r of matrix is $r = n$ then the columns are independent.

$r < n \rightarrow$ dependence

Span

The vectors $v_1 - - - v_n$ span a space S then S consists of all linear combinations of S.

$\Rightarrow$ The smallest space which contains all the vectors $v_1 - - - v_n$ is the span of these vectors

Basis → A basis for a vector space is a sequence of vectors $v_1 \ldots v_d$ →

- They are independent.
- They span the vector space.

Eg $\mathbb{R}^3$

→ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ I $N(I) = 0$

→ $\begin{bmatrix} 1 & 1 & 4 \\ 2 & 1 & 9 \\ 3 & 1 & -5 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 2 & 5 & 8 \end{bmatrix}$$

X

③ = 2 × ② - ①

Now to check if the cols are independent we do elimination & find the rank of a matrix.

- For a square matrix $\underbrace{n \times n}_{\mathbb{R}^2}$, it must be **invertible**

imp $\boxed{\text{"dimension" of space = \# of vectors}}$

And all the basis for a given space have equal number of vectors.

- The number of vectors in the basis of a span is its <u>dimension</u>.

eg $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$

- span is the <u>$C(A)$</u>

- They are not a basis as they are dependent.

- $N(A)$ is not just zero

eg $\begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}$ is <u>in $N(A)$</u>

- Basis for $C(A) \rightarrow$ cols 1 & 2

- Basis of $C(A)$ can be found by taking the pivot cols.

$\underline{\text{rank}}\ (A) = \underline{\#\text{ pivot cols}} = \underline{\text{dimension}\,(C(A))}$

$$\text{dimension}\,(N(A)) = [\,n - \text{rank}\,(A),\ 1\,]$$

for given ex $\boxed{\dim\,(N(A)) = 2}$

$\underline{n - \text{rank}(A)} = \#\ \underline{\text{free vars}} = \underline{\dim\,(N(A))}$